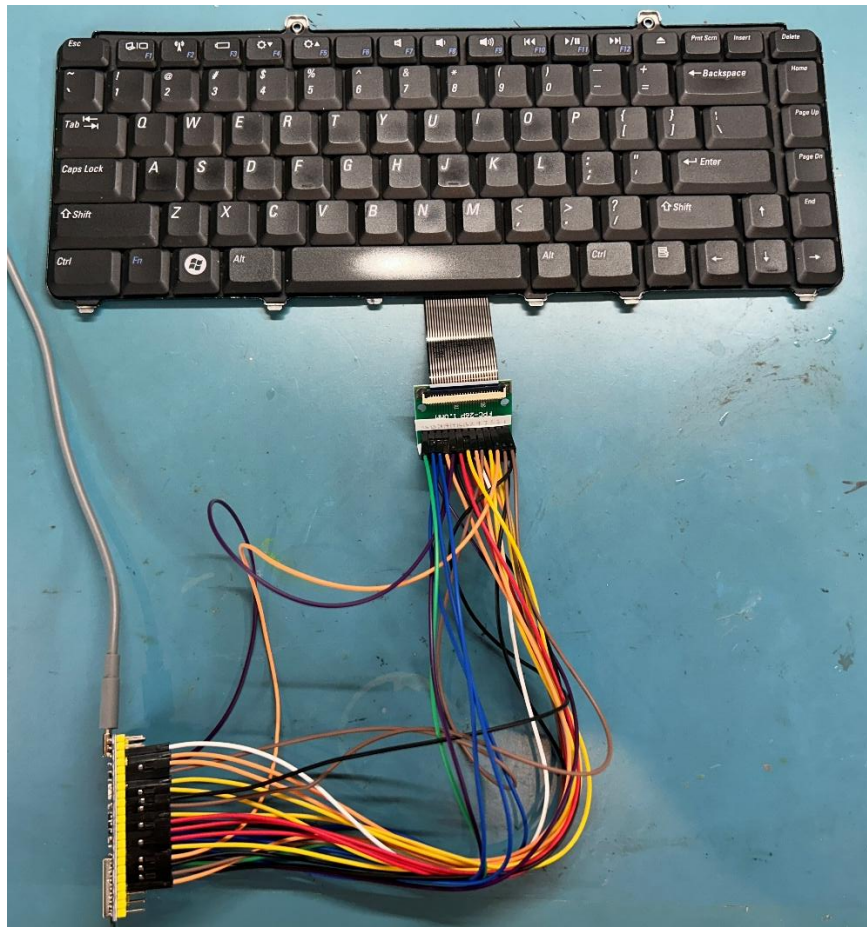
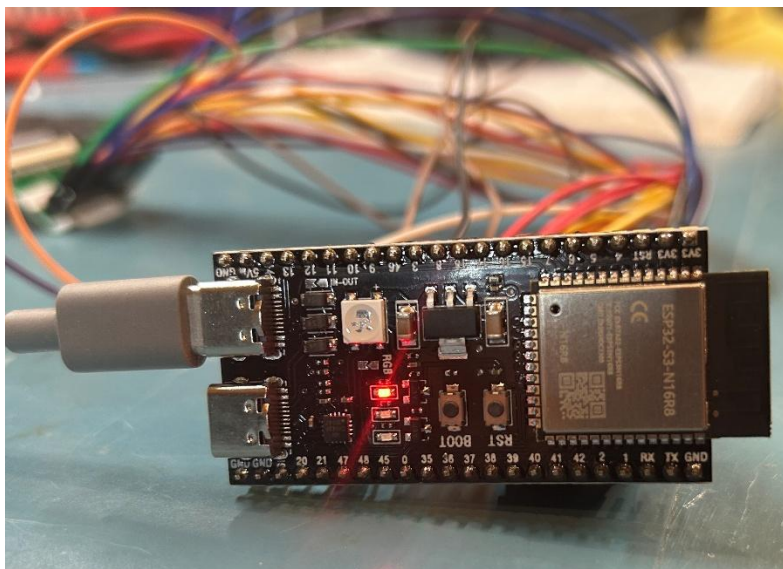


ESP32-S3 – KMK Keyboard Controller

A Dell 1545 keyboard connected to an FPC adapter board with jumper wires to an [ESP32-S3](#)



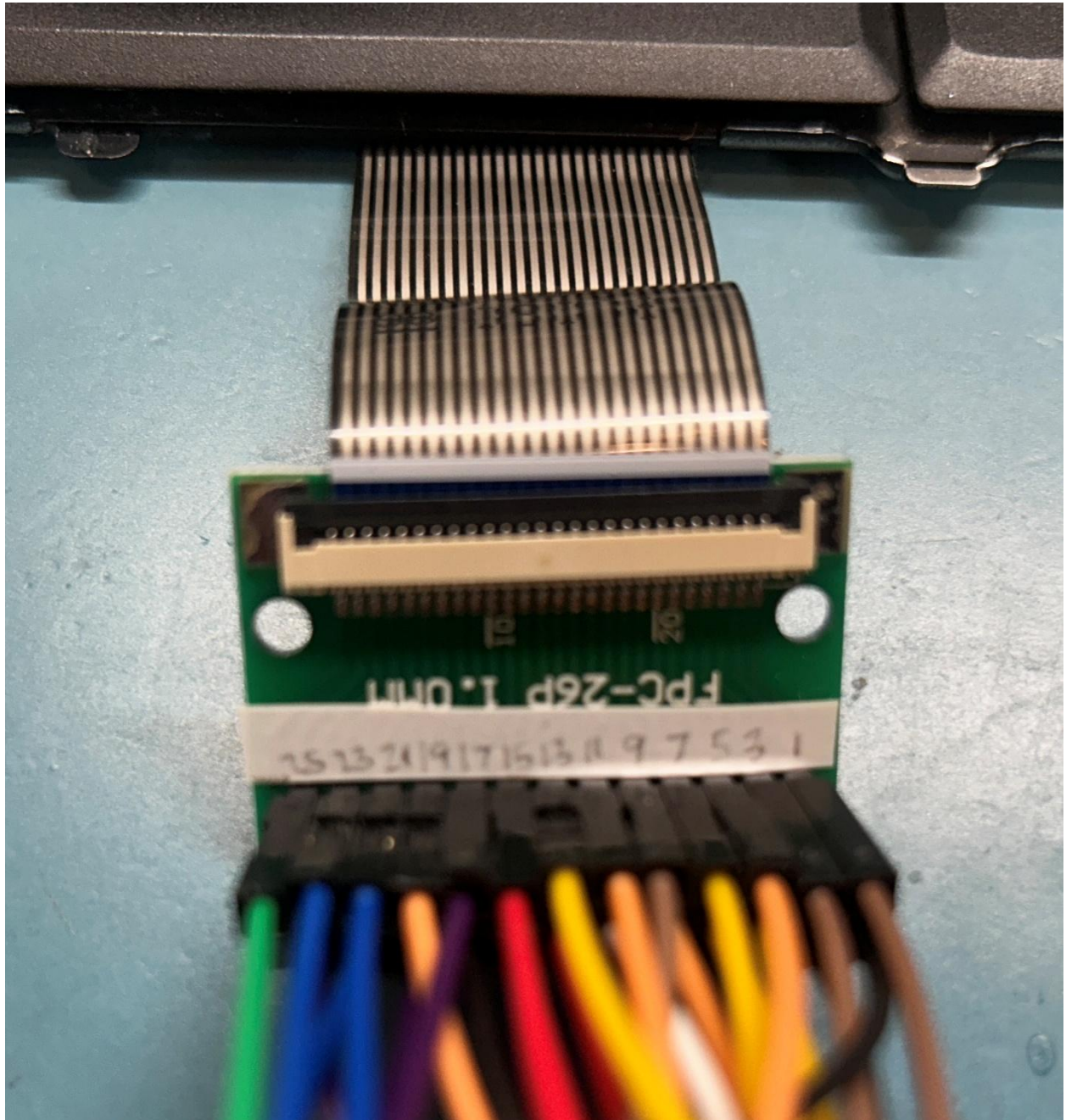
ESP32-S3 N16R8



FPC to GPIO Jumper Table

FPC	GPIO
1	3
2	46
3	5
4	6
5	7
6	8
7	9
8	10
9	11
10	12
11	13
12	14
13	15
14	16
15	17
16	18
17	21
18	1
19	4
20	38
21	39
22	40
23	41
24	42
25	2

I covered the silkscreen numbers on the adapter board and renumbered with pin 1 starting on the right side instead of the left.



The procedure to load CircuitPython and the KMK library is the same as documented in step 16 of my [Instructable](#).

The KMK code.py for this project is given at my [repo](#).

A few differences from my Pi Pico code are:

The “import microcontroller” line was needed for row and column naming and they had to be named “microcontroller.pin.GPIO” followed by the number.

After connecting the 23 GPIO’s that are totally unused, I needed 2 more so I used GPIO3 and 46. These must be used as inputs so I chose ROW2COL for the diode orientation to make the rows inputs and the columns outputs.

The Dell 1545 Key Matrix is given below. FPC number first, followed by GPIO number

FPC-IO	18-1	19-4	20-38	21-39	22-40	23-41	24-42	25-2
1-3	KC.NO	KC.INS	KC.NO	KC.F12	KC.NO	KC.NO	KC.NO	KC.RGHT
2-46	KC.NO	KC.DEL	KC.NO	KC.F11	KC.NO	KC.NO	KC.NO	KC.DOWN
3-5	KC.UP	KC.HOME	KC.APP	KC.END	KC.NO	KC.NO	KC.NO	KC.LEFT
4-6	KC.NO	KC.F8	KC.F7	KC.N9	KC.O	KC.L	KC.DOT	KC.NO
5-7	KC.QUOT	KC.MINS	KC.LBRC	KC.N0	KC.P	KC.SCLN	KC.NO	KC.SLSH
6-8	KC.F6	KC.EQL	KC.RBRC	KC.N8	KC.I	KC.K	KC.COMM	KC.NO
7-9	KC.H	KC.N6	KC.Y	KC.N7	KC.U	KC.J	KC.M	KC.N
8-10	KC.F5	KC.F9	KC.BSPC	KC.F10	KC.NO	KC.BSLS	KC.ENT	KC.SPC
9-11	KC.G	KC.N5	KC.T	KC.N4	KC.R	KC.F	KC.V	KC.B
10-12	KC.F4	KC.F2	KC.F3	KC.N3	KC.E	KC.D	KC.C	KC.NO
11-13	KC.NO	KC.F1	KC.CAPS	KC.N2	KC.W	KC.S	KC.X	KC.NO
12-14	KC.ESC	KC.GRV	KC.TAB	KC.N1	KC.Q	KC.A	KC.Z	KC.NO
13-15	KC.LALT	KC.NO	KC.NO	KC.PSCR	KC.NO	KC.NO	KC.NO	KC.RALT
14-16	KC.NO	KC.NO	KC.LSFT	KC.NO	KC.NO	KC.NO	KC.RSFT	KC.NO
15-17	KC.NO	KC.LCTL	KC.NO	KC.NO	KC.NO	KC.NO	KC.RCTL	KC.NO
16-18	KC.NO	KC.NO	KC.NO	KC.LGUI	KC.PGUP	KC.PGDN	KC.NO	KC.NO
17-21	KC.NO	KC.NO	KC.NO	KC.NO	KC.NO	FN	KC.NO	KC.NO

Conclusion: The ESP32-S3 works as a KMK keyboard controller but there are GPIO pin limitations. Only 23 GPIO are completely unused by on-board circuitry. The remainder are used in some way such as GPIO 3 and 46 which can only be used as inputs. I had to use these pins to get the required 25 I/O pins for the Dell 1545 keyboard. Luckily, I had already mapped out this keyboard so I knew which FPC pins could be designated as inputs but that’s not the case for an unknown key matrix.